

Probabilistic graphical models

1 Markov Networks and Conditional Random Fields (Lecture 6)

This lecture covers the parameterization of Markov networks (MNs), their log-linear form, conditional random fields (CRFs), and how MNs and Bayesian networks (BNs) relate to and convert into one another.

1.1 Independencies of a Markov network (recap)

A trail $X_1 - \dots - X_n$ is *active* given Z if none of the X_i (other than the endpoints) is in Z ; unlike in a BN, activity of a trail in an MN depends only on the graph structure, not on the direction of edges or on colliders. This gives three families of independence statements associated with an MN $\mathcal{H}$:

- **Global Markov:** $\mathcal{I}(\mathcal{H}) = \{(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) : \text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})\}$, i.e. $\mathbf{X}$ and $\mathbf{Y}$ are independent given $\mathbf{Z}$ whenever every trail between them is blocked by $\mathbf{Z}$.
- **Pairwise Markov:** $\mathcal{I}_p(\mathcal{H}) = \{(X \perp Y \mid \mathcal{X} - \{X, Y\}) : X - Y \notin \mathcal{H}\}$, any two non-adjacent nodes are independent given everything else.
- **Local Markov:** $\mathcal{I}_\ell(\mathcal{H}) = \{(X \perp \mathcal{X} - \{X\} - \text{MB}_{\mathcal{H}}(X) \mid \text{MB}_{\mathcal{H}}(X)) : X \in \mathcal{X}\}$, a node is independent of the rest of the graph given its Markov blanket (here, just its neighbors).

These three notions coincide for positive distributions; they are stated in decreasing order of strength (global $\Rightarrow$ pairwise $\Rightarrow$ local in general, with equivalence under positivity).

1.2 Gibbs distribution and alternative parameterizations

The MN's graph being an I-map for P is equivalent to P factorizing over the graph's cliques:

$$P(\mathbf{X}) = \frac{1}{Z} \prod_{i=1}^m \phi_i(\mathbf{D}_i), \quad Z = \sum_{\mathbf{x} \in \text{Val}(\mathbf{X})} \prod_{i=1}^m \phi_i(\mathbf{d}_i),$$

where each $\mathbf{D}_i$ is (a subset of) a clique and Z is the *partition function*, a normalizing constant. Writing $U(\mathbf{X}) = \prod_i \phi_i(\mathbf{D}_i)$, $P(\mathbf{X}) = U(\mathbf{X})/Z$ is called a *Gibbs distribution*.

1.2.1 Logarithmic / energy representation

Since factors are non-negative, each can be written as $\phi_i(\mathbf{D}_i) = e^{-\psi_i(\mathbf{D}_i)}$ for some $\psi_i : \text{Val}(\mathbf{D}_i) \rightarrow \mathbb{R}$. Substituting,

$$P(\mathbf{X}) = \frac{1}{Z} \exp\left(-\sum_i \psi_i(\mathbf{D}_i)\right), \quad \ln P(\mathbf{X}) = -\ln Z - \sum_i \psi_i(\mathbf{D}_i).$$

The quantity $\sum_i \psi_i(\mathbf{D}_i)$ is called the *energy*: lower energy corresponds to higher probability. A *feature* is a function $f : \text{Val}(\mathbf{D}_i) \rightarrow \mathbb{R}$ that need not be non-negative (a factor without the positivity requirement); a canonical example is an *indicator feature* that is 1 for a particular assignment and 0 otherwise (e.g. “these two superpixels have the same color”).

Worked example. For the 4-cycle $A-B-C-D-A$ with pairwise factors $\phi_1(A, B)$, $\phi_2(B, C)$, $\phi_3(C, D)$, $\phi_4(D, A)$, set $\psi_i = -\ln \phi_i$ (so $\phi_1(a^0, b^0) = 30 \Rightarrow \psi_1(a^0, b^0) = -\ln 30 = -3.4$, etc.). Then

$$P(A, B, C, D) \propto \exp \left[-\psi_1(A, B) - \psi_2(B, C) - \psi_3(C, D) - \psi_4(D, A) \right].$$

1.3 Log-linear Markov networks

A *log-linear* MN is specified by a set of features $\mathcal{F} = \{f_1(\mathbf{D}_1), \dots, f_k(\mathbf{D}_k)\}$ (each with scope $\mathbf{D}_i$, a complete subgraph) and weights $w_1, \dots, w_k$:

$$P(X_1, \dots, X_n) = \frac{1}{Z} \exp \left[-\sum_{i=1}^k w_i f_i(\mathbf{D}_i) \right], \quad \ln P(X_1, \dots, X_n) \propto \sum_{i=1}^k w_i f_i(\mathbf{D}_i).$$

Different features may share a scope, and features/weights can be reused across different factors (typically: features hand-designed, weights learned from data). Log-linear form does not change the independence assertions of the underlying model; it only changes how factors are parameterized. Log-probability is linear in the weights, which is what makes maximum-likelihood weight learning a (concave) convex optimization problem.

Example. With $f_1(\text{Smoking}, \text{Cancer}) = \mathbb{I}[\neg \text{Smoking} \vee \text{Cancer}]$ and $w_1 = 1.5$, the feature/weight pair encodes a soft implication “smoking tends to cause cancer” as one term in $\sum_i w_i f_i$.

1.3.1 Example: Ising model

Binary spins $X_i \in \{+1, -1\}$ on a grid, with pairwise interaction potentials $\phi_{i,j}(x_i, x_j) = w_{i,j} x_i x_j$ between neighbors and single-node bias potentials $\phi_i(x_i) = u_i x_i$:

$$P(\xi) = \frac{1}{Z} \exp \left(-\sum_{i < j} w_{i,j} x_i x_j - \sum_i u_i x_i \right).$$

1.3.2 Example: Boltzmann machine

A (general) *Boltzmann machine* is a fully-connected, pairwise MN (Ising-like) over binary units $\mathbf{s} \in \{0, 1\}^n$, split into *visible* units $\mathbf{v}$ (which carry the data) and *hidden* units $\mathbf{h}$ (which carry latent structure). Every pair of units – visible-visible, hidden-hidden, and visible-hidden – may be connected. Its energy is a general quadratic form

$$E(\mathbf{v}, \mathbf{h}) = -\frac{1}{2} \mathbf{v}^\top L \mathbf{v} - \frac{1}{2} \mathbf{h}^\top J \mathbf{h} - \mathbf{v}^\top W \mathbf{h} - \mathbf{c}^\top \mathbf{v} - \mathbf{b}^\top \mathbf{h},$$

with $P(\mathbf{v}, \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{v}, \mathbf{h}))$, $Z = \sum_{\mathbf{v}, \mathbf{h}} \exp(-E(\mathbf{v}, \mathbf{h}))$. This is just a log-linear MN whose features are all pairwise products $s_i s_j$ and all singletons s_i , i.e. a fully general pairwise binary MRF. Because the graph is a complete graph (every unit is a neighbor of every other unit), none of the units are conditionally independent of each other given the rest, so exact inference and even computing Z are intractable in general (the sum has $2^{|\mathbf{v}|+|\mathbf{h}|}$ terms); this is the price paid for the extra representational power of hidden units and dense connectivity.

Restricted Boltzmann machine (RBM). Restricting the graph to be *bipartite* – dropping all visible-visible edges ($L = 0$) and all hidden-hidden edges ($J = 0$), keeping only visible-hidden edges W – gives the RBM:

$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{v}^\top W \mathbf{h} - \mathbf{c}^\top \mathbf{v} - \mathbf{b}^\top \mathbf{h}, \quad P(\mathbf{v}, \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{v}, \mathbf{h})).$$

The point of removing intra-layer edges is exactly the pairwise-Markov property from Section 1.1: with no visible-visible or hidden-hidden edges, each visible unit is separated from the other visible units by the hidden layer (and vice versa), so

$$P(\mathbf{h} \mid \mathbf{v}) = \prod_j P(h_j \mid \mathbf{v}), \quad P(\mathbf{v} \mid \mathbf{h}) = \prod_i P(v_i \mid \mathbf{h}),$$

i.e. all hidden units are conditionally independent given the visible units, and all visible units are conditionally independent given the hidden units. Expanding the energy for a single unit shows each conditional is a logistic/sigmoid unit exactly as in Section 2.2 (CRFs generalize logistic regression):

$$P(h_j = 1 \mid \mathbf{v}) = \text{sigmoid} \left(b_j + \sum_i v_i W_{ij} \right), \quad P(v_i = 1 \mid \mathbf{h}) = \text{sigmoid} \left(c_i + \sum_j W_{ij} h_j \right).$$

This is what makes RBMs practical: although Z (and hence $P(\mathbf{v}, \mathbf{h})$ itself) is still intractable to compute exactly, one can alternate sampling $\mathbf{h} \mid \mathbf{v}$ and $\mathbf{v} \mid \mathbf{h}$ in *block Gibbs sampling*, since each block factorizes into independent coordinates. This alternating structure underlies contrastive-divergence training, and stacking RBMs (feeding one layer's hidden samples as the next RBM's visible data) is the basis of deep belief networks.

1.3.3 Metric MRFs

A special case of pairwise MRFs used for labeling problems (image segmentation, denoising): all x_i take values in a common label space V , and neighbors are encouraged to take similar labels via a smoothness energy

$$E(x_1, \dots, x_n) = \sum_i \epsilon_i(x_i) + \sum_{(i,j) \in \mathcal{E}} \epsilon_{i,j}(x_i, x_j), \quad P \propto \exp(-E),$$

where $\epsilon_{i,j}$ is built from a distance/metric function μ on labels, e.g. $\epsilon(x_i, x_j) = \min(c \|x_i - x_j\|_p, \text{dist}_{\max})$ (a robust, truncated distance). Inference reduces to minimizing the energy, equivalently maximizing the probability.

2 Conditional Random Fields (CRFs)

2.1 Motivation

Often we only care about $P(\mathbf{Y} \mid \mathbf{X})$ for target variables $\mathbf{Y}$ given observed variables $\mathbf{X}$, not the full joint $P(\mathbf{X}, \mathbf{Y})$. Modeling $P(\mathbf{X}, \mathbf{Y})$ directly would force us to also model the (possibly complicated, high-dimensional, or continuous) distribution over $\mathbf{X}$ itself. A CRF instead defines an MN over $\{\mathbf{X} \cup \mathbf{Y}\}$ with factors $\phi_1(\mathbf{D}_1), \dots, \phi_m(\mathbf{D}_m)$, but normalizes only over $\mathbf{Y}$, for each fixed value of $\mathbf{X}$:

$$P(\mathbf{Y} \mid \mathbf{X}) = \frac{1}{Z(\mathbf{X})} \tilde{P}(\mathbf{Y}, \mathbf{X}), \quad \tilde{P}(\mathbf{Y}, \mathbf{X}) = \prod_{i=1}^m \phi_i(\mathbf{D}_i), \quad Z(\mathbf{X}) = \sum_{\mathbf{Y}} \tilde{P}(\mathbf{Y}, \mathbf{X}).$$

This is parameterized exactly like a Gibbs distribution, but normalized differently (per- $\mathbf{X}$ rather than globally). Consequences:

- We never need to model dependencies among the X 's, which may be complex, poorly understood, continuous, or simply not of interest.
- We are free to use highly expressive, even overlapping/correlated features of $\mathbf{X}$ (e.g. many overlapping windows of text or image patches) without worrying about violating an independence assumption on $\mathbf{X}$ – unlike, say, Naive Bayes, which assumes features are conditionally independent given the label and breaks down when features (e.g. neighboring pixels or words) are highly correlated.
- CRFs are widely used in NLP (sequence/entity labeling) and image segmentation.

2.2 CRFs generalize logistic regression

Logistic regression is the special case of a conditional MN with a single binary query variable Y and binary features $X_1, \dots, X_k$. With pairwise potentials $\phi_i(X_i, Y) = \exp\{w_i \mathbb{I}[X_i = 1, Y = 1]\}$ and a node potential $\phi_0(Y) = \exp\{w_0 \mathbb{I}[Y = 1]\}$,

$$\tilde{P}(Y=1 \mid x_1, \dots, x_k) = \exp \left\{ w_0 + \sum_i w_i x_i \right\}, \quad \tilde{P}(Y=0 \mid \cdot) = 1,$$

so that, after normalizing,

$$P(Y=1 \mid x_1, \dots, x_k) = \text{sigmoid} \left(w_0 + \sum_{i=1}^k w_i x_i \right),$$

the familiar logistic regression model. This shows logistic regression is a CRF with trivial (star-shaped) structure, and motivates CRFs as its generalization to structured, correlated $\mathbf{Y}$.

2.3 Applications

- **Image segmentation:** one node per superpixel X_i , edges between adjacent regions, a class label S_i per superpixel. Because everything is conditioned on the (given) image, correlation among pixels does not need to be modeled; node/edge factors can use arbitrary image features (color histograms, texture, CNN outputs). Modern architectures (CRF-as-RNN, dense CRFs stacked on deep nets) embed CRF inference as a layer trained end-to-end with a CNN, rather than as a disconnected post-processing step.
- **Text/sequence labeling** (e.g. named entity recognition): target labels Y_t (e.g. B-PER/I-PER/B-LOC/I-LOC/OTH) chained together with pairwise factors $\Phi_1(Y_t, Y_{t+1})$ enforcing label consistency, and per-position factors $\Phi_2(Y_t, X_1, \dots, X_T)$ tying each label to (features of) the whole observed sequence.
- **Graph convolutional networks:** a CRF layer can be inserted between convolutional layers to encourage connected/similar nodes to have similar hidden representations, i.e. the CRF acts as a regularizer on the network's hidden features rather than on raw observations.

3 Bayesian Networks vs. Markov Networks

3.1 Relative expressiveness

- **Directed (BN):** captures inter-causal/“explaining away” reasoning; parameters (CPDs) are individually interpretable and can be elicited by hand; easy to generate data by ancestral

sampling; large body of work on latent-variable models.

- **Undirected (MN)**: captures symmetric “affinity” relationships; naturally represents cyclic dependency structures; parameters are less interpretable and usually learned from data; trickier (but tractable) parameter estimation; easy to add overlapping factors/features.

Neither family strictly contains the other: some independence patterns (e.g. induced by a single common cause vs. a 4-cycle of pairwise dependencies) can be captured exactly by a BN but not any MN, and vice versa.

3.2 BN $\rightarrow$ MN: moralization

Every factor of a BN (i.e. every CPD, a node together with its parents) must correspond to a subset of some clique of the target MN. **Moralize** the graph: drop edge directions and add an (undirected) edge between any two nodes that share a child (“marrying the parents”), so that every node-plus-parents set becomes a clique. The moralized graph is a *minimal I-map* for the BN; it is a perfect map only if the original DAG had no immoralities (v-structures) to begin with. Moralization can lose independence information that was present in the original DAG (extra edges = fewer independencies).

3.3 MN $\rightarrow$ BN: triangulation

Given an MN $\mathcal{H}$, build a minimal I-map BN by fixing an elimination/variable ordering (each X_i before its descendants) and, for $i = 1, \dots, n$, setting $\text{Parents}(X_i)$ to the minimal subset $\mathbf{S} \subseteq \{X_1, \dots, X_{i-1}\}$ such that $X_i \perp (\{X_1, \dots, X_{i-1}\} \setminus \mathbf{S}) \mid \mathbf{S}$ in $\mathcal{H}$. Applied along an ordering that visits the MN’s cycle structure, this typically introduces extra directed edges (to avoid immoralities) – the resulting BN must be **chordal** (every cycle of length ≥ 4 has a chord), since a v-structure-free DAG cannot contain chordless long cycles. Equivalently, converting MN $\rightarrow$ BN requires **triangulating** the MN’s graph (adding edges until it is chordal); different elimination orderings can produce very different numbers of added edges, and the conversion can lose independence information.

3.4 Perfect maps and chordal graphs

Moralize (BN $\rightarrow$ MN) and triangulate (MN $\rightarrow$ BN) are, in general, lossy in opposite directions. The distributions representable *perfectly* (as a P-map) in both families are exactly those whose MN is an **undirected chordal graph** (a graph whose minimal cycles have length ≤ 3). **Theorem:** if $\mathcal{H}$ is non-chordal, there is no BN G with $I(G) = I(\mathcal{H})$ – because a minimal I-map BN for $\mathcal{H}$ must itself be chordal, so representing a non-chordal MN exactly as a BN is impossible without adding edges that destroy some of its independence assertions.

4 Factor Graphs

A **factor graph** is a third graphical representation (alongside BNs and MNs) that makes the factorization of a distribution explicit, rather than only its independence structure. It is bipartite, with two kinds of node:

- *variable nodes* (drawn as circles), one per random variable;
- *factor nodes* (drawn as squares), one per factor ϕ , connected to exactly the variables in its scope.

Unlike an MN graph – which only records *that* some factor exists over each clique, collapsing all factors sharing that clique into one edge set – a factor graph distinguishes individual factors, including multiple factors over the same or overlapping scopes (e.g. separate pairwise factors ϕ_1, ϕ_2, ϕ_3 plus higher-order factors ϕ_4 over triangle $\{X, Y, Z\}$, or extra unary factors ϕ_5, ϕ_6, ϕ_7 on individual variables). This makes factor graphs the natural representation for deriving and running message-passing (sum-product / belief propagation) inference algorithms, since messages are exchanged along the explicit variable–factor edges.